

Supplementary Material for:

“Let Time Tell: Identification and Gaussian Process Estimation for Interrupted Time Series”

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A Proof of Theorem 1

Proof of Theorem 1. Fix a unit i and a post-treatment period $t^* \geq t_0$. By consistency (1), it holds that $Y_{it^*} = Y_{it^*}(1)$. The target estimand (7) can therefore be written as

$$\tau_{it^*} = Y_{it^*}(1) - \mathbb{E}[Y_{it^*}(0) \mid \mathbf{X}_{it^*}] = Y_{it^*} - \mathbb{E}[Y_{it^*}(0) \mid \mathbf{X}_{it^*}] = Y_{it^*} - g_i(\mathbf{X}_{it^*}), \quad (\text{A.1})$$

so it remains to identify $g_i(\mathbf{X}_{it^*}) = \mathbb{E}[Y_{it^*}(0) \mid \mathbf{X}_{it^*}]$ from the observed data.

The counterfactual conditional mean coincides with g_i by definition. By the structural model (3) and the linearity of conditional expectation,

$$\begin{aligned} \mathbb{E}[Y_{it^*}(0) \mid \mathbf{X}_{it^*}] &= \mathbb{E}[h_i(\mathbf{X}_{it^*}, \mathbf{U}_{it^*}) + \varepsilon_{it^*} \mid \mathbf{X}_{it^*}] \\ &= \mathbb{E}[h_i(\mathbf{X}_{it^*}, \mathbf{U}_{it^*}) \mid \mathbf{X}_{it^*}] + \mathbb{E}[\varepsilon_{it^*} \mid \mathbf{X}_{it^*}] \\ &= \int_{\mathcal{U}} h_i(\mathbf{X}_{it^*}, \mathbf{u}) dP(\mathbf{U}_{it^*} = \mathbf{u} \mid \mathbf{X}_{it^*}) = g_i(\mathbf{X}_{it^*}), \end{aligned} \quad (\text{A.2})$$

where $\mathbb{E}[\varepsilon_{it^*} \mid \mathbf{X}_{it^*}] = \mathbb{E}[\mathbb{E}[\varepsilon_{it^*} \mid \mathbf{X}_{it^*}, \mathbf{U}_{it^*}] \mid \mathbf{X}_{it^*}] = 0$ by iterated expectations applied to (3), and the last equality is the definition (4). Identity (A.2) holds for any model of the form (3), with or without Assumption 1; it does not identify $g_i(\mathbf{X}_{it^*})$, since its right-hand side depends on the unobserved distribution $P(\mathbf{U}_{it^*} \mid \mathbf{X}_{it^*})$ and at t^* only $Y_{it^*} = Y_{it^*}(1)$ is observed.

Assumption 1 supplies the link. For each t it states $\mathbb{E}[Y_{it}(0) \mid \mathbf{X}_{it}, \mathbf{U}_{it}] = \mathbb{E}[Y_{it}(0) \mid \mathbf{X}_{it}]$; the left-hand side equals $h_i(\mathbf{X}_{it}, \mathbf{U}_{it})$ and the right-hand side equals $g_i(\mathbf{X}_{it})$, so

$$h_i(\mathbf{X}_{it}, \mathbf{U}_{it}) = g_i(\mathbf{X}_{it}) \quad \text{a.s.,} \quad \forall t. \quad (\text{A.3})$$

Because h_i enters (3) with a form not depending on t (Section 2), the g_i in (A.3) is common across periods, so the systematic part of $Y_{it}(0)$ depends on $\mathbf{X}_{it}$ alone and is identical before and after t_0 .

It remains to recover g_i from pre-treatment data. For $t < t_0$, consistency (1) and no anticipation (2) give $Y_{it} = Y_{it}(0)$, so by (3) and (A.3),

$$Y_{it} = h_i(\mathbf{X}_{it}, \mathbf{U}_{it}) + \varepsilon_{it} = g_i(\mathbf{X}_{it}) + \varepsilon_{it}, \quad \mathbb{E}[\varepsilon_{it} \mid \mathbf{X}_{it}] = 0, \quad t < t_0. \quad (\text{A.4})$$

This is a nonparametric regression of the observed Y_{it} on $\mathbf{X}_{it}$ with regression function g_i , so g_i is identified on the support of $\mathbf{X}_{i,\text{pre}}$. Since (A.3) holds at t^* with the same g_i , evaluating this function

at $\mathbf{X}_{it^*}$ returns $g_i(\mathbf{X}_{it^*}) = \mathbb{E}[Y_{it^*}(0) \mid \mathbf{X}_{it^*}]$ by (A.2), and substituting into (A.1) identifies τ_{it^*} . $\square$

The pre-treatment regression identifies g_i on the support of the observed inputs. Because time is included among the covariates, evaluating g_i at a post-treatment input requires extrapolation beyond that support. Section B characterizes the resulting uncertainty over admissible counterfactual continuations.

B Connections to RKHS Theory

This appendix establishes the theoretical results of Section 4. We suppress the unit subscript i throughout. Proposition 1 follows by taking conditional expectations in the four-term decomposition (19) under its stated assumptions. We first prove Proposition 2, then develop its noiseless and minimax implications and the refinements associated with kernel choice.

B.1 RKHS Preliminaries

Let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a positive definite kernel. Its reproducing kernel Hilbert space $\mathcal{H}_k$ is the completion of $\text{span}\{k(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{X}\}$ under the inner product determined by

$$\langle k(\cdot, \mathbf{x}), k(\cdot, \mathbf{x}') \rangle_{\mathcal{H}_k} = k(\mathbf{x}, \mathbf{x}'), \quad (\text{B.1})$$

for all $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$. This inner product yields the reproducing property: for any $f \in \mathcal{H}_k$,

$$f(\mathbf{x}) = \langle f, k(\cdot, \mathbf{x}) \rangle_{\mathcal{H}_k}, \quad (\text{B.2})$$

so point evaluation is the inner product of f with $k(\cdot, \mathbf{x})$. Applying Cauchy–Schwarz to (B.2) gives

$$|f(\mathbf{x})| \leq \|f\|_{\mathcal{H}_k} \sqrt{k(\mathbf{x}, \mathbf{x})}, \quad (\text{B.3})$$

which establishes that point evaluation is a bounded linear functional on $\mathcal{H}_k$ and underlies the worst-case bound of Proposition 2.

The RKHS norm $\|f\|_{\mathcal{H}_k}$ measures complexity relative to the kernel, whereas the $L^2(P_{\mathbf{X}})$ norm measures average squared magnitude under the distribution of $\mathbf{X}$. The distinction matters here

because the former restricts the admissible function class, while the latter defines the best approximation g^* in (18). The best approximation g^* in (18) minimizes the $L^2(P_{\mathbf{X}})$ distance to g over the RKHS ball $\mathcal{F}_B$. For a bounded continuous kernel on a compact domain, the norm-bounded ball has a compact image in $L^2(P_{\mathbf{X}})$, ensuring the existence of a minimizer. The budget also guarantees $\|g^*\|_{\mathcal{H}_k} \leq B$, as required for the worst-case bound.

The pre-treatment inputs span a finite-dimensional subspace of $\mathcal{H}_k$,

$$\mathcal{S}_{\text{pre}} = \text{span}\{k(\cdot, \mathbf{X}_1), \dots, k(\cdot, \mathbf{X}_{t_0-1})\}. \quad (\text{B.4})$$

By the representer theorem (Kimeldorf and Wahba, 1970), the minimizer of the regularized empirical risk over $\mathcal{H}_k$ lies in $\mathcal{S}_{\text{pre}}$, and for squared loss with regularization σ^2 the GP posterior mean takes the form

$$\hat{g} = \sum_{t=1}^{t_0-1} w_t k(\cdot, \mathbf{X}_t), \quad \mathbf{w} = (\mathbf{K}_{\text{pre}} + \sigma^2 \mathbf{I})^{-1} \mathbf{Y}_{\text{pre}}, \quad (\text{B.5})$$

recovering the kernel ridge regression estimator (Rasmussen et al., 2006, Ch. 6). The regularization parameter (σ^2) penalizes the RKHS norm, while the representer theorem confines ($\hat{g}$) to the span of the pre-treatment kernel features. The component of (g^*) that these features cannot reconstruct is central to the learning-error bound proved next.

B.2 Proof of Proposition 2

We prove Proposition 2 stated in Section 4 of the main text. The proof follows results in Kanagawa et al. (2025), adapted to our setting.

Proposition B.1 (Proposition 2 (GP posterior variance controls the learning error)). *Suppose g lies in the unit ball $\mathcal{F}_1$, so that $g^* = g$, and suppose outcomes are observed with noise, $Y_t = g^*(\mathbf{X}_t) + \varepsilon_t$, where ε_t are mean-zero with variance $\sigma^2 > 0$. The GP posterior mean is the weighting estimator $\hat{g}(\mathbf{X}_{t^*}) = \sum_t w_t Y_t$ with weights $\mathbf{w} = (\mathbf{K}_{\text{pre}} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_{t^*}$, and its posterior variance is*

$$\mathbb{V}_{t^*} = k(\mathbf{X}_{t^*}, \mathbf{X}_{t^*}) - \mathbf{k}_{t^*}^\top (\mathbf{K}_{\text{pre}} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_{t^*}.$$

Let $k_{t^*}^\perp = k(\cdot, \mathbf{X}_{t^*}) - \sum_t w_t k(\cdot, \mathbf{X}_t)$ be the residual of the test feature after its ridge reconstruction from the training-feature span $\mathcal{S}_{\text{pre}} = \text{span}\{k(\cdot, \mathbf{X}_t)\}$, coinciding with the orthogonal-projection

residual onto $\mathcal{S}_{\text{pre}}$ as $\sigma^2 \rightarrow 0$. Then the learning error decomposes as

$$g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*}) = \underbrace{\left(g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t) \right)}_{\text{deterministic error}} - \underbrace{\sum_t w_t \varepsilon_t}_{\text{noise propagation}},$$

and its two sources are controlled by the two summands of the GP posterior variance:

$$\left| g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t) \right| \leq \|k_{t^*}^\perp\|_{\mathcal{H}_k}, \quad \text{Var}\left[\sum_t w_t \varepsilon_t\right] = \sigma^2 \|\mathbf{w}\|^2, \quad \underbrace{\|k_{t^*}^\perp\|_{\mathcal{H}_k}^2 + \sigma^2 \|\mathbf{w}\|^2}_{\text{GP posterior variance}} = \mathbb{V}_{t^*}.$$

Therefore, the first summand of $\mathbb{V}_{t^*}$ gives the squared worst-case bound on the deterministic error and the second gives the variance of the propagated noise.

Proof of Proposition B.1. Let $\mathbf{w} = (\mathbf{K}_{\text{pre}} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_{t^*}$ denote the posterior mean weights, where $\mathbf{k}_{t^*} = (k(\mathbf{X}_1, \mathbf{X}_{t^*}), \dots, k(\mathbf{X}_{t_0-1}, \mathbf{X}_{t^*}))^\top$. By (B.5), $\hat{g}(\mathbf{X}_{t^*}) = \sum_t w_t Y_t$. Recall the residual feature

$$k_{t^*}^\perp = k(\cdot, \mathbf{X}_{t^*}) - \sum_{t=1}^{t_0-1} w_t k(\cdot, \mathbf{X}_t), \quad (\text{B.6})$$

the residual of the test feature after the ridge reconstruction. As $\sigma^2 \rightarrow 0$, the ridge reconstruction becomes the orthogonal projection onto $\mathcal{S}_{\text{pre}}$, and $\|k_{t^*}^\perp\|_{\mathcal{H}_k}$ measures the distance of the test feature from that subspace. Corollary B.1 treats this limiting case.

By the reproducing property,

$$g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t) = \langle g^*, k_{t^*}^\perp \rangle_{\mathcal{H}_k}. \quad (\text{B.7})$$

Applying Cauchy–Schwarz:

$$\left| g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t) \right| \leq \|g^*\|_{\mathcal{H}_k} \cdot \|k_{t^*}^\perp\|_{\mathcal{H}_k}. \quad (\text{B.8})$$

It remains to bound $\|k_{t^*}^\perp\|_{\mathcal{H}_k}$. Expanding via the reproducing property and bilinearity:

$$\|k_{t^*}^\perp\|_{\mathcal{H}_k}^2 = k(\mathbf{X}_{t^*}, \mathbf{X}_{t^*}) - 2\mathbf{k}_{t^*}^\top \mathbf{w} + \mathbf{w}^\top \mathbf{K}_{\text{pre}} \mathbf{w}. \quad (\text{B.9})$$

Substituting $\mathbf{K}_{\text{pre}} = (\mathbf{K}_{\text{pre}} + \sigma^2 \mathbf{I}) - \sigma^2 \mathbf{I}$ and using the definition of $\mathbf{w}$:

$$\mathbf{w}^\top \mathbf{K}_{\text{pre}} \mathbf{w} = \mathbf{k}_{t^*}^\top (\mathbf{K}_{\text{pre}} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_{t^*} - \sigma^2 \|\mathbf{w}\|^2. \quad (\text{B.10})$$

Since $\mathbf{k}_{t^*}^\top \mathbf{w} = \mathbf{k}_{t^*}^\top (\mathbf{K}_{\text{pre}} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_{t^*}$, combining terms yields

$$\|k_{t^*}^\perp\|_{\mathcal{H}_k}^2 = \underbrace{k(\mathbf{X}_{t^*}, \mathbf{X}_{t^*}) - \mathbf{k}_{t^*}^\top (\mathbf{K}_{\text{pre}} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_{t^*}}_{\mathbb{V}[f(\mathbf{X}_{t^*}) \mid \mathbf{X}_{\text{pre}}, Y_{\text{pre}}]} - \underbrace{\sigma^2 \|\mathbf{w}\|^2}_{\geq 0}. \quad (\text{B.11})$$

Combining (B.8) with (B.11) and $\|g^*\|_{\mathcal{H}_k} \leq 1$ gives the deterministic bound $|g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t)| \leq \|k_{t^*}^\perp\|_{\mathcal{H}_k}$, while the noise propagation $\sum_t w_t \varepsilon_t$ is mean-zero with variance $\sigma^2 \|\mathbf{w}\|^2$. Rearranging (B.11) gives $\|k_{t^*}^\perp\|_{\mathcal{H}_k}^2 + \sigma^2 \|\mathbf{w}\|^2 = \mathbb{V}_{t^*}$, so the two summands of the posterior variance are exactly the two scales just bounded, and in particular $\|k_{t^*}^\perp\|_{\mathcal{H}_k} \leq \sqrt{\mathbb{V}_{t^*}}$. This completes the proof. $\square$

Under kernel misspecification the GP posterior variance can understate the true prediction error (Beckers et al., 2018), but the worst-case bound remains valid for any $g^* \in \mathcal{H}_k$. Figure B.1 illustrates the decomposition.

The noiseless case isolates the geometric leg as a result of its own.

Corollary B.1 (Noiseless training). *Suppose the pre-treatment outcomes are observed without noise, $Y_t = g^*(\mathbf{X}_t)$ for $t < t_0$, with $\mathbf{K}_{\text{pre}}$ invertible. In the limit $\sigma^2 \rightarrow 0$ of Proposition 2, the weights become the interpolation weights $\mathbf{w} = \mathbf{K}_{\text{pre}}^{-1} \mathbf{k}_{t^*}$, the residual $k_{t^*}^\perp$ is the orthogonal projection residual of $k(\cdot, \mathbf{X}_{t^*})$ onto $\mathcal{S}_{\text{pre}}$, and the learning error is deterministic with*

$$|g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*})| \leq \|k_{t^*}^\perp\|_{\mathcal{H}_k} = \sqrt{\mathbb{V}_{t^*}}, \quad (\text{B.12})$$

with equality attained at $g^* \propto k_{t^*}^\perp$.

Setting $\sigma^2 = 0$ eliminates the noise-propagation term, leaving $\mathbb{V}_{t^*} = \|k_{t^*}^\perp\|_{\mathcal{H}_k}^2$. The remaining bound therefore reflects only what the pre-treatment inputs cannot reconstruct.

Proposition 2 is sharp over the unit ball rather than merely an upper bound, since $\|k_{t^*}^\perp\|_{\mathcal{H}_k} = \sup_{\|g^*\|_{\mathcal{H}_k} \leq 1} |\langle g^*, k_{t^*}^\perp \rangle_{\mathcal{H}_k}|$, attained at $g^* \propto k_{t^*}^\perp$. Figure B.2 reads this in function space, drawing a family of least favorable targets that meet the band, each at one horizon.

Minimax Optimality at the Extrapolation Point. The preceding results bound the GP's worst-case pointwise error. We now ask whether another estimator using the same pre-treatment data could guarantee a smaller error. Under noiseless training, let $D(g) = (g(\mathbf{X}_1), \dots, g(\mathbf{X}_{t_0-1}))$.

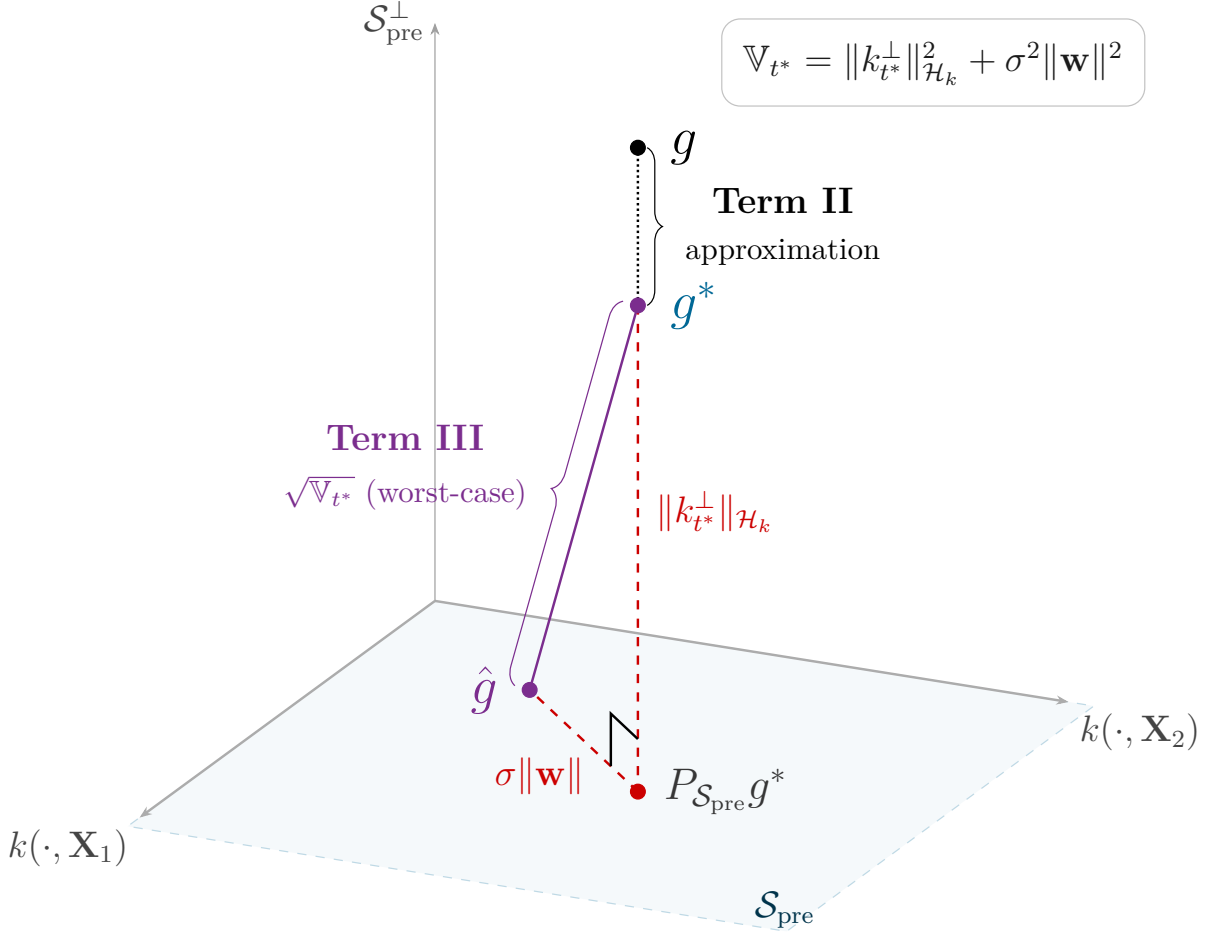


Figure B.1. Schematic of the error decomposition at t^* in $\mathcal{H}_k$. *Note:* The plane is $\mathcal{S}_{\text{pre}} = \text{span}\{k(\cdot, \mathbf{X}_1), \dots, k(\cdot, \mathbf{X}_{t_0-1})\}$. The true CEF g lies outside $\mathcal{H}_k$, its best RKHS approximation g^* is estimated by $\hat{g} \in \mathcal{S}_{\text{pre}}$, the gap $g - g^*$ is the approximation error (Term II), and $g^* - \hat{g}$ is the learning error (Term III), which splits into a component orthogonal to $\mathcal{S}_{\text{pre}}$ and a noise component within it. Over $\|g^*\|_{\mathcal{H}_k} \leq 1$ the geometric leg is longest for the target aligned with $k_{t^*}^\perp$, drawn here with length $\|k_{t^*}^\perp\|_{\mathcal{H}_k}$; a generic target gives the smaller $\langle g^*, k_{t^*}^\perp \rangle$. With the noise leg $\sigma\|\mathbf{w}\|$ the two legs combine into the hypotenuse, $\sqrt{\mathbb{V}_{t^*}}$.

Allowing any estimator $\varphi : \mathbb{R}^{t_0-1} \rightarrow \mathbb{R}$, define the minimax risk at the extrapolation point as

$$\mathfrak{R}(\mathbf{X}_{t^*}) = \inf_{\varphi} \sup_{\|g\|_{\mathcal{H}_k} \leq 1} |g(\mathbf{X}_{t^*}) - \varphi(D(g))|. \quad (\text{B.13})$$

The infimum searches over all estimators; the supremum searches over all regression functions in the unit ball.

Proposition B.2 (Minimax optimality of the noiseless band). *Under the conditions of Corol-*

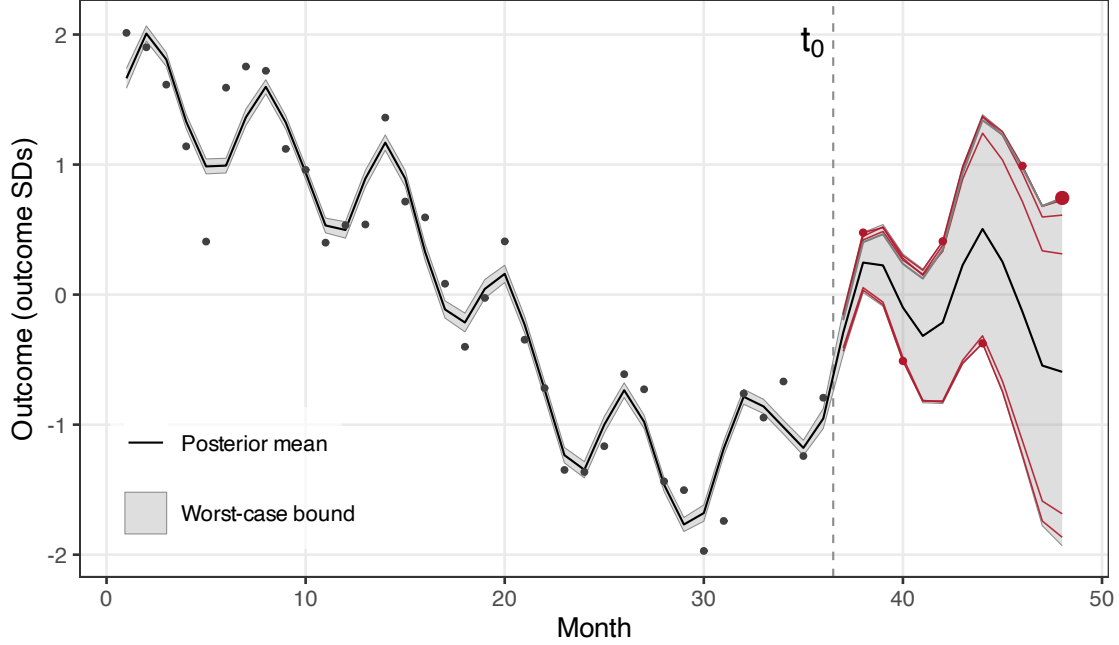


Figure B.2. Attainment in Proposition 2, in function space. The shaded band is the geometric term $\|k_t^\perp\|_{\mathcal{H}_k}$ of the posterior variance, the largest error any counterfactual with $\|g^*\|_{\mathcal{H}_k} \leq 1$ agreeing with the pre-treatment series can incur at t^* , so it is the attained worst case rather than a loose envelope, exactly under noiseless training (Corollary B.1) and approximately under regularization. The bold red curve is the least favorable admissible counterfactual at the furthest horizon, meeting the band edge at t^* (point); the thin curves meet it at their own horizons ($t^* = 2, 4, 6, 8, 10$), so the bound is attained pointwise across the window.

lary B.1, with $\|k_{t^*}^\perp\|_{\mathcal{H}_k} > 0$,

$$\mathfrak{R}(\mathbf{X}_{t^*}) = \|k_{t^*}^\perp\|_{\mathcal{H}_k} = \sqrt{\mathbb{V}_{t^*}},$$

and the GP posterior mean $\hat{g}(\mathbf{X}_{t^*}) = \mathbf{k}_{t^*}^\top \mathbf{K}_{\text{pre}}^{-1} \mathbf{Y}_{\text{pre}}$ attains the infimum. Over the ball of budget B the risk scales to $B \|k_{t^*}^\perp\|_{\mathcal{H}_k}$, attained by the same estimator.

Proof. Lower bound. Set $g_\pm = \pm k_{t^*}^\perp / \|k_{t^*}^\perp\|_{\mathcal{H}_k}$. Both have unit norm, so both lie in the unit ball. Because $k_{t^*}^\perp$ is orthogonal to $\mathcal{S}_{\text{pre}}$, both functions vanish at every pre-treatment input: $D(g_+) = D(g_-) = \mathbf{0}$. Any estimator/mapping φ therefore returns the same value $d = \varphi(\mathbf{0})$ for both. Yet at the extrapolation point the two functions separate. By the reproducing property and the orthogonal decomposition $k(\cdot, \mathbf{X}_{t^*}) = k_{t^*}^\perp + P_{\mathcal{S}_{\text{pre}}} k(\cdot, \mathbf{X}_{t^*})$,

$$g_\pm(\mathbf{X}_{t^*}) = \frac{\langle k_{t^*}^\perp, k(\cdot, \mathbf{X}_{t^*}) \rangle_{\mathcal{H}_k}}{\pm \|k_{t^*}^\perp\|_{\mathcal{H}_k}} = \pm \|k_{t^*}^\perp\|_{\mathcal{H}_k}.$$

The two values sit at $\pm \|k_{t^*}^\perp\|_{\mathcal{H}_k}$, so for every $d \in \mathbb{R}$,

$$\max\{|g_+(\mathbf{X}_{t^*}) - d|, |g_-(\mathbf{X}_{t^*}) - d|\} \geq \|k_{t^*}^\perp\|_{\mathcal{H}_k},$$

and the floor $\mathfrak{R}(\mathbf{X}_{t^*}) \geq \|k_{t^*}^\perp\|_{\mathcal{H}_k}$ follows.

Upper bound. Corollary B.1 states that the GP posterior mean satisfies $|g(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*})| \leq \|k_{t^*}^\perp\|_{\mathcal{H}_k}$ for every g in the unit ball. Its worst case therefore meets the floor, and the two bounds coincide. The budget- B statement follows by replacing g with g/B . The argument adapts the classical theory of optimal recovery (Golomb and Weinberger, 1959; Micchelli and Rivlin, 1977) to the extrapolation target of the ITS design. $\square$

The lower bound uses two functions that produce identical pre-treatment evaluations but disagree at the extrapolation point. The upper bound holds over the entire unit ball, showing that the GP posterior mean attains the smallest possible worst-case pointwise error in the noiseless setting.

Data-conditional sharpening. The unconditional minimax risk $\|k_{t^*}^\perp\|_{\mathcal{H}_k}$ is the worst case across all datasets. With a specific observed series Y_{pre} , the argument tightens. Let $\hat{g}$ be the minimum-norm interpolant of Y_{pre} , satisfying $\|\hat{g}\|_{\mathcal{H}_k} \leq 1$. Any admissible function g with $D(g) = Y_{\text{pre}}$ and $\|g\|_{\mathcal{H}_k} \leq 1$ decomposes as $g = \hat{g} + u$ with $u \perp \mathcal{S}_{\text{pre}}$. By the Pythagorean identity, $\|u\|_{\mathcal{H}_k}^2 \leq 1 - \|\hat{g}\|_{\mathcal{H}_k}^2$, so u has less budget to spend in the unseen direction. The reproducing property gives $g(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*}) = \langle u, k_{t^*}^\perp \rangle_{\mathcal{H}_k}$, and applying Cauchy–Schwarz yields the identified set

$$\{g(\mathbf{X}_{t^*}) : \|g\|_{\mathcal{H}_k} \leq 1, D(g) = Y_{\text{pre}}\} = [\hat{g}(\mathbf{X}_{t^*}) \pm \sqrt{1 - \|\hat{g}\|_{\mathcal{H}_k}^2} \|k_{t^*}^\perp\|_{\mathcal{H}_k}],$$

with the endpoints attained at $u \propto \pm k_{t^*}^\perp$. The observed pre-treatment series uses part of the RKHS budget, leaving less room for functions that agree on the observed inputs but differ at t^* . The resulting set is centered at the minimum-norm interpolant and is contained in the unconditional worst-case band.

Noise does not restore learnability. Let $q_{t^*} = k(\cdot, \mathbf{X}_{t^*}) - P_{\mathcal{S}_{\text{pre}}}k(\cdot, \mathbf{X}_{t^*})$ denote the orthogonal-projection residual. The least favorable pair $g_\pm = \pm q_{t^*}/\|q_{t^*}\|_{\mathcal{H}_k}$ agrees at every pre-treatment input and therefore induces identical distributions of the noisy observations, provided

the noise law is the same under both functions. No estimator can distinguish this pair from the training data. The same two-point argument consequently gives

$$\sup_{\|g\|_{\mathcal{H}_k} \leq 1} \mathbb{E} |g(\mathbf{X}_{t^*}) - \varphi(Y_{\text{pre}})| \geq \|q_{t^*}\|_{\mathcal{H}_k}.$$

Additional pre-treatment observations can reduce this lower bound by enlarging the training-feature span, but noise alone cannot identify directions orthogonal to that span.

B.3 Kernel Choice, Source Conditions, and Extrapolation

This section examines how kernel choice and target alignment affect the RKHS norm and the worst-case learning bound. We first consider the cost of representing trends, then characterize the combined kernel and the refinement obtained under a source condition. We also distinguish these finite-sample extrapolation bounds from conventional in-support convergence results.

B.3.1 Spectral Representation of the RKHS Norm

Let $P_{\mathbf{X}}$ be the marginal distribution of $\mathbf{X}$ and $T_k: L^2(P_{\mathbf{X}}) \rightarrow L^2(P_{\mathbf{X}})$ the kernel integral operator $T_k(f)(\mathbf{x}) = \int k(\mathbf{x}, \mathbf{x}') f(\mathbf{x}') dP_{\mathbf{X}}(\mathbf{x}')$. On a compact domain with a continuous kernel, T_k is compact, positive, and self-adjoint, with Mercer decomposition

$$k(\mathbf{x}, \mathbf{x}') = \sum_{j \geq 1} \mu_j \phi_j(\mathbf{x}) \phi_j(\mathbf{x}'), \quad (\text{B.14})$$

where $\mu_1 \geq \mu_2 \geq \dots > 0$ are eigenvalues and $\{\phi_j\}$ are eigenfunctions orthonormal in $L^2(P_{\mathbf{X}})$ (Steinwart and Christmann, 2008). The RKHS norm then admits the representation

$$\|f\|_{\mathcal{H}_k}^2 = \sum_{j \geq 1} \frac{f_j^2}{\mu_j}, \quad (\text{B.15})$$

with $f_j = \langle f, \phi_j \rangle_{L^2(P_{\mathbf{X}})}$ (Kanagawa et al., 2025). Energy on eigenfunctions with large eigenvalues costs little, while energy on high-frequency eigenfunctions, those with small μ_j , pays a steep penalty.

A stationary kernel carries a trend only by spreading it across many small-eigenvalue directions, and (B.15) charges for each. The reproducing property fixes the total charge without computing the spectrum. Take $g(t) = \beta t$ over a window of length T and let h approximate it to within ε in

the supremum norm. Applying (B.2) and Cauchy–Schwarz to the difference of two feature maps,

$$\|h\|_{\mathcal{H}_{\text{Gaussian}}} \geq \frac{|h(T) - h(0)|}{\sqrt{2}} \geq \frac{|\beta| T - 2\varepsilon}{\sqrt{2}}, \quad (\text{B.16})$$

since $\|k(\cdot, T) - k(\cdot, 0)\|_{\mathcal{H}_k}^2 = 2 - 2k(0, T) \leq 2$ for a unit-diagonal kernel, while h must climb at least $|\beta|T - 2\varepsilon$ across the window. The cost of a trend is the distance it climbs, and no bandwidth avoids it: the bound reads the kernel only through its diagonal.

For a fixed approximation tolerance ε , the bound in (B.16) grows with the length of the window. The combined kernel avoids this cost for linear trends by representing βt through its linear component, where $\|\beta t\|_{\mathcal{H}_{\text{lin}}}^2 = \beta^2$, independently of the window length. The kernel-sum identity in Section B.3.2 formalizes this advantage.

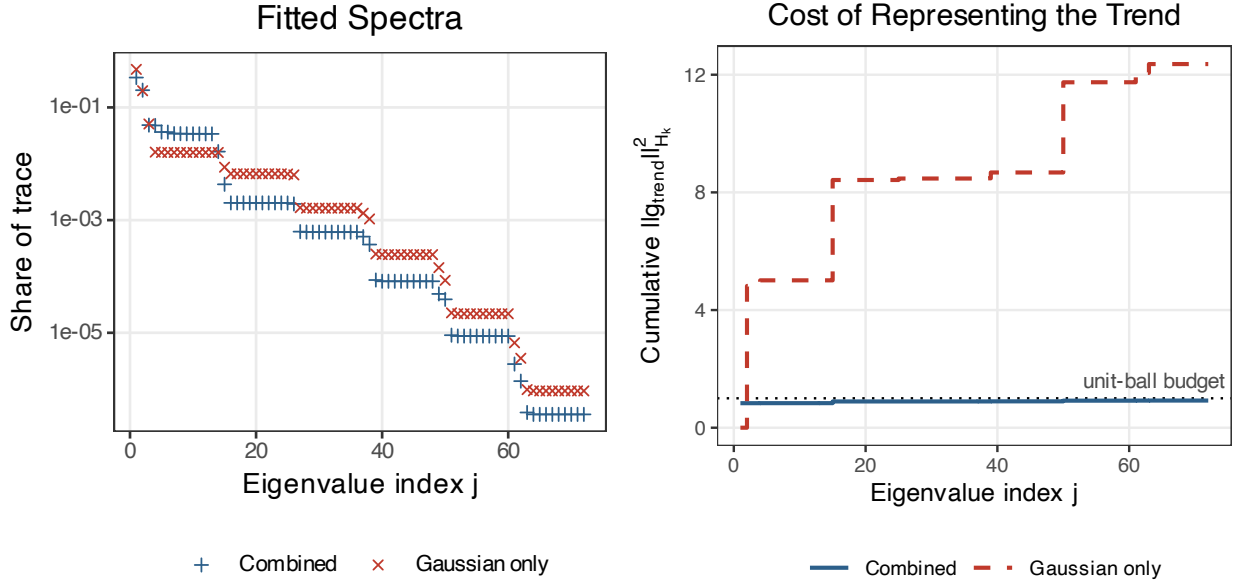


Figure B.3. What the kernel sum buys, on the national *Heller* pre-treatment series ($n = 72$ months). Left panel plots the eigenvalues of each fitted kernel matrix as shares of its trace on a logarithmic scale, the prior variance the kernel assigns to each orthogonal direction on the design. Right panel accumulates over those same directions the squared RKHS norm a unit-variance trend costs, with the dotted line marking the unit-ball budget of Proposition 2. Under the combined kernel the trend is the leading direction, carrying thirty-four percent of the trace, and costs 0.924 with ninety percent of that paid in the leading direction alone. Under the Gaussian-only kernel the leading direction is orthogonal to the trend, which is assembled instead from many small directions at a cost of 12.4. In norm the budgets are $B = 0.96$ and $B = 3.52$.

For a single kernel the cost of a trend can be bounded analytically, as in (B.16); for a sum the

norm is an infimum over decompositions (B.18) and no such bound is available, so we read the fitted spectrum instead. Figure B.3 does this on the national *Heller* pre-treatment series. On an equally spaced monthly design the Gaussian kernel matrix is unchanged by reversing time, so each eigenvector is either symmetric or antisymmetric about the midpoint; the leading one is symmetric and the centered trend is antisymmetric, leaving the two orthogonal at any bandwidth. The trend is then assembled from many small directions, each charged by (B.15), and the budget reaches $B = 3.52$, in line with the lower bound of (B.16). The linear component supplies the missing direction, and the budget falls to $B = 0.96$, inside the unit ball. Universality is not what separates the two specifications, since a Gaussian kernel does represent the trend on this window; the price of doing so is.

B.3.2 RKHS of Kernel Sums

Theorem B.1 (Aronszajn 1950). *For positive definite kernels $k_1, \dots, k_m$ on $\mathcal{X}$, the RKHS of $k = k_1 + \dots + k_m$ is*

$$\mathcal{H}_k = \{f_1 + \dots + f_m : f_j \in \mathcal{H}_{k_j}\}, \quad (\text{B.17})$$

with norm

$$\|f\|_{\mathcal{H}_k}^2 = \inf \left\{ \sum_{j=1}^m \|f_j\|_{\mathcal{H}_{k_j}}^2 : f = \sum_{j=1}^m f_j, f_j \in \mathcal{H}_{k_j} \right\}. \quad (\text{B.18})$$

The infimum is attained. When the component RKHSs intersect only at $\{0\}$, the decomposition is unique and $\mathcal{H}_k$ is an orthogonal direct sum.

The infimum assigns each function the cost of its cheapest decomposition. For our kernel $k = k_{\text{Gaussian}} + k_{\text{per}} + k_{\text{lin}}$ the component RKHSs overlap on a compact domain, so the decomposition is not unique and the infimum is no larger than any particular feasible decomposition.

Suppose $g \in \mathcal{H}_k$ with decomposition $g = g_{\text{lin}} + g_{\text{per}} + g_{\text{Gaussian}}$. Theorem B.1 yields the bound

$$\|g\|_{\mathcal{H}_k}^2 \leq \|g_{\text{lin}}\|_{\mathcal{H}_{\text{lin}}}^2 + \|g_{\text{per}}\|_{\mathcal{H}_{\text{per}}}^2 + \|g_{\text{Gaussian}}\|_{\mathcal{H}_{\text{Gaussian}}}^2, \quad (\text{B.19})$$

which follows by dropping the infimum in (B.18) over any feasible decomposition.

The linear kernel norm $\|g_{\text{lin}}\|_{\mathcal{H}_{\text{lin}}}^2 = \|\beta\|^2$ depends only on the slope and not on the horizon, the periodic norm depends on the seasonal amplitude and smoothness, and the Gaussian norm is

moderate because g_{Gaussian} is a stationary residual the Gaussian kernel represents efficiently. By contrast, a Gaussian-only specification pays a norm that grows with the window's length for the same target (Section B.3.1, Figure B.3).

B.3.3 Source Condition

The source condition (Caponnetto and De Vito, 2007) characterizes kernel–target alignment more finely than membership by quantifying how fast the eigencoefficients of g^* decay relative to the kernel's eigenvalues.

Definition B.1 (Source condition). *Let T_k be the kernel integral operator with Mercer eigensystem $\{(\mu_j, \phi_j)\}_{j \geq 1}$, and define $T_k^r f = \sum_{j \geq 1} \mu_j^r f_j \phi_j$ for $r \geq 0$, with $f_j = \langle f, \phi_j \rangle_{L^2(P_{\mathbf{X}})}$. The best RKHS approximation g^* satisfies a source condition of order r if there exists $v \in \mathcal{H}_k$ with*

$$g^* = T_k^r v. \quad (\text{B.20})$$

The case $r = 0$ reduces to ordinary RKHS membership, as assumed in Proposition 2. As r increases, the condition places greater weight on the kernel's leading spectral directions and restricts the target to functions more closely aligned with them. This restriction concerns the learning target g^* and is separate from the approximation error in Term II.

In the fitted combined kernel, the leading direction captures the standardized trend. Figure B.4 shows how increasing r suppresses local movements associated with smaller eigenvalues, leaving functions more closely aligned with that trend. Alignment thus depends on both the target function and the kernel used to represent it.

Proposition B.3 (Norm control under the source condition). *If $g^* = T_k^r v$ for some $v \in \mathcal{H}_k$ and $r \geq 0$, then*

$$\|g^*\|_{\mathcal{H}_k}^2 \leq \mu_1^{2r} \|v\|_{\mathcal{H}_k}^2. \quad (\text{B.21})$$

Proof. Since $g^* = T_k^r v$, its L^2 eigencoefficients are $(g^*)_j = \mu_j^r v_j$ with $v_j = \langle v, \phi_j \rangle_{L^2(P_{\mathbf{X}})}$. By (B.15),

$$\|g^*\|_{\mathcal{H}_k}^2 = \sum_{j \geq 1} \frac{(\mu_j^r v_j)^2}{\mu_j} = \sum_{j \geq 1} \mu_j^{2r-1} (v_j)^2 \leq \mu_1^{2r} \sum_{j \geq 1} \frac{(v_j)^2}{\mu_j} = \mu_1^{2r} \|v\|_{\mathcal{H}_k}^2,$$

where the inequality uses $\mu_j \leq \mu_1$. □

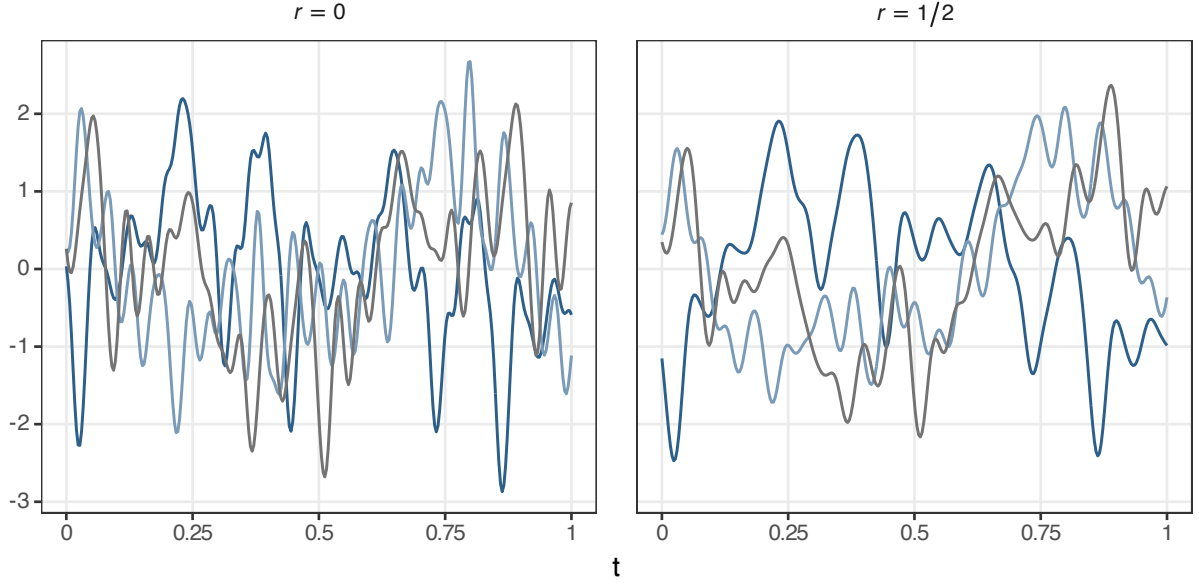


Figure B.4. Illustration of the source condition using the fitted combined kernel. Each panel shows three draws $g = T_k^r v$, computed from its empirical eigensystem and rescaled to unit sample variance for display. The same three v , drawn uniformly from the unit sphere of $\mathcal{H}_k$, are used in both panels. The left panel takes $r = 0$, while the right takes $r = 1/2$. Increasing r dampens directions associated with smaller eigenvalues, reducing local variation.

The ratio $\|g^*\|_{\mathcal{H}_k}^2 / \|v\|_{\mathcal{H}_k}^2 = \sum_j \mu_j^{2r} (v_j^2 / \mu_j) / \|v\|_{\mathcal{H}_k}^2$ is a weighted average of μ_j^{2r} over the spectral directions in which v carries energy, with μ_1^{2r} as the worst case. The source condition concentrates the target on directions the kernel represents cheaply, controlling the norm factor $\|g^*\|_{\mathcal{H}_k}$ in Proposition 2; Proposition B.4 below controls the geometric factor $\|k_{t^*}^\perp\|_{\mathcal{H}_k}$ separately. Under the standard normalization $\mu_1 \leq 1$ the factor μ_1^{2r} is a genuine reduction; for an unnormalized composite kernel it quantifies spectral alignment rather than unconditional norm reduction.

The source condition refines the worst-case learning bias of Proposition 2 directly rather than only through the norm factor, since over the source class of order r the worst-case bias is governed by a smoothed residual.

Proposition B.4 (Alignment-refined worst-case learning bias). *Suppose g^* satisfies the source condition of order $r \geq 0$ (Definition B.1), $g^* = T_k^r v$ with $\|v\|_{\mathcal{H}_k} \leq 1$. Then the deterministic learning bias of Proposition 2 satisfies*

$$\left| g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t) \right| = |\langle g^*, k_{t^*}^\perp \rangle_{\mathcal{H}_k}| \leq \|T_k^r k_{t^*}^\perp\|_{\mathcal{H}_k}, \quad (\text{B.22})$$

and the right-hand side is the exact worst case as g^* ranges over $\{T_k^r v : \|v\|_{\mathcal{H}_k} \leq 1\}$. Writing $a_j = \langle k_{t^*}^\perp, \sqrt{\mu_j} \phi_j \rangle_{\mathcal{H}_k}$ and $\nu_j = \mu_j/\mu_1 \in (0, 1]$,

$$\|T_k^r k_{t^*}^\perp\|_{\mathcal{H}_k}^2 = \mu_1^{2r} \sum_{j \geq 1} \nu_j^{2r} a_j^2 \leq \mu_1^{2r} \|k_{t^*}^\perp\|_{\mathcal{H}_k}^2, \quad (\text{B.23})$$

with equality iff $k_{t^*}^\perp$ is supported on the leading eigenspace. At $r = 0$ the bound equals the constant $\|k_{t^*}^\perp\|_{\mathcal{H}_k}$ of Proposition 2.

Proof. The first equality is the representer identity used for Proposition 2. In the orthonormal basis $\{\sqrt{\mu_j} \phi_j\}_{j \geq 1}$ of $\mathcal{H}_k$ the operator T_k^r is diagonal with entries μ_j^r , hence self-adjoint on $\mathcal{H}_k$, so $\langle T_k^r v, k_{t^*}^\perp \rangle_{\mathcal{H}_k} = \langle v, T_k^r k_{t^*}^\perp \rangle_{\mathcal{H}_k}$. Maximizing over $\|v\|_{\mathcal{H}_k} \leq 1$ gives $\|T_k^r k_{t^*}^\perp\|_{\mathcal{H}_k}$, attained at $v \propto T_k^r k_{t^*}^\perp$. Expanding $k_{t^*}^\perp = \sum_j a_j \sqrt{\mu_j} \phi_j$ and applying T_k^r coordinatewise yields (B.23); the inequality uses $\nu_j \leq 1$. $\square$

Membership in the combined $\mathcal{H}_k$ holds the norm factor $\|g^*\|_{\mathcal{H}_k}$ finite and horizon-free (Section B.3.2). A source condition of order $r > 0$ tightens the geometric factor: by (B.23), $\|T_k^r k_{t^*}^\perp\|_{\mathcal{H}_k}^2 = \mu_1^{2r} \sum_j \nu_j^{2r} a_j^2$ with $\nu_j = \mu_j/\mu_1 \in (0, 1]$, so $\sum_j \nu_j^{2r} a_j^2 \leq \sum_j a_j^2 = \|k_{t^*}^\perp\|_{\mathcal{H}_k}^2$ is a strict reduction whenever $k_{t^*}^\perp$ has energy outside the leading eigenspace, regardless of whether $\mu_1 \leq 1$.

B.3.4 In-Support Convergence and Extrapolation

Standard kernel learning theory provides convergence rates for estimation on the support of the training distribution under appropriate sampling, regularization, and spectral conditions (Caponnetto and De Vito, 2007; Kanagawa et al., 2025). These results concern average estimation accuracy within the training distribution. They do not, however, establish accuracy at post-treatment inputs beyond the observed pre-treatment support.

Our estimation procedure also may not choose its hyperparameters to satisfy the rate-optimal tuning conditions of these results. We therefore rely on the finite-sample, pointwise bound of Proposition 2 to characterize learning error at extrapolation points. The next section examines the remaining uncertainty directly, by asking how far admissible counterfactual functions can disagree after treatment despite agreeing on the pre-treatment inputs.

B.4 Extrapolation Uncertainty of the Model Class

This section derives the closed-form expression for the extrapolation uncertainty defined in Definition 1 and relates it to the noiseless minimax risk.

Corollary B.2 (Closed form of the extrapolation uncertainty). *Under the conditions of Corollary B.1, with $k_{t^*}^\perp$ the orthogonal projection residual,*

$$\omega_B(\mathbf{X}_{t^*}) = 2B \|k_{t^*}^\perp\|_{\mathcal{H}_k} = 2B \sqrt{\mathbb{V}_{t^*}}. \quad (\text{B.24})$$

Proof. Write $u = f - f'$ for an admissible pair. Then $\|u\|_{\mathcal{H}_k} \leq 2B$ by the triangle inequality, and u vanishes at every pre-treatment input, so $u \perp \mathcal{S}_{\text{pre}}$ by the reproducing property. Conversely, every $u \perp \mathcal{S}_{\text{pre}}$ with $\|u\|_{\mathcal{H}_k} \leq 2B$ arises from the admissible pair $(u/2, -u/2)$. Hence

$$\omega_B(\mathbf{X}_{t^*}) = \sup\{|\langle u, k_{t^*}^\perp \rangle_{\mathcal{H}_k}| : \|u\|_{\mathcal{H}_k} \leq 2B, u \perp \mathcal{S}_{\text{pre}}\} = 2B \|k_{t^*}^\perp\|_{\mathcal{H}_k},$$

where the first equality uses $u(\mathbf{X}_{t^*}) = \langle u, k(\cdot, \mathbf{X}_{t^*}) \rangle_{\mathcal{H}_k} = \langle u, k_{t^*}^\perp \rangle_{\mathcal{H}_k}$ for $u \perp \mathcal{S}_{\text{pre}}$. The second equality follows from Cauchy–Schwarz and is attained by $u = 2B k_{t^*}^\perp / \|k_{t^*}^\perp\|_{\mathcal{H}_k}$ when the residual is nonzero. If the residual is zero, both sides are zero. $\square$

The closed form shows that extrapolation uncertainty is twice the noiseless minimax risk over the same RKHS ball (Proposition B.2). It measures the largest disagreement permitted by the kernel and budget among functions that agree at all pre-treatment inputs. Conditioning on a particular noiseless pre-treatment series can further restrict this set, as shown in Section B.2. With noisy observations, the geometric uncertainty remains, while Proposition 2 separately accounts for noise propagated through the GP prediction.

C Simulation Details

This section records the data-generating processes for the calibrated simulation of Section 5 and connects each scenario to the error decomposition of Section 4.

C.1 Common Structure

We generate monthly outcomes $y_t = g(t) + \varepsilon_t$ for $t = 1, \dots, n_{\text{pre}} + 12$, with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, treatment beginning at $t_0 = n_{\text{pre}} + 1$, a fixed twelve-month post-treatment horizon, and a constant effect τ added over that horizon. We write z_t for the time index standardized on the pre-treatment window, and the annual period is twelve. The three scenarios are designed around monthly data with seasonality, as in the empirical application. The two trend-based scenarios also include a small random Gaussian bump,

$$r(t) = a \exp\{-(t - \mu)^2 / (2w^2)\}, \quad a \sim U(-0.3, 0.3), \mu \sim U(8, n_{\text{pre}} - 4), w \sim U(4, 8),$$

that displaces each trend from a clean closed form. The bump introduces transient local deviations of the kind an applied series would carry, so the estimator cannot exploit exact knowledge of the parametric shape. Throughout, $\beta \sim U(0.8, 1.3)$, $A \sim U(0.7, 1)$, $\phi \sim U(0, 2\pi)$, $\sigma^2 = 0.05$ and $\tau = 0.6$. The noise variance and treatment effect are held fixed across scenarios so that the comparison focuses on differences in the untreated trajectories.

C.2 Three Simulation Scenarios

The three scenarios vary the shape of the untreated trajectory, from a kernel-based function to nonlinear trends with different seasonal components. This allows us to examine how coverage changes when the generating trajectory is less closely aligned with the fitted GP.

Smooth. A kernel-based random function:

$$g(t) = 1.5 \bar{f}(t), \quad f(t) = \sum_{j=1}^{10} \omega_j k(z_t, z_j), \quad \omega_j \sim \mathcal{N}(0, 1), \quad z_j \sim U(z_1, z_{n_{\text{pre}}+12}).$$

Here $\bar{f}$ denotes the pre-treatment standardization of f , and k is the Gaussian+Periodic+Linear kernel of Section 3.2 at a fixed length-scale. The finite kernel expansion makes this a favorable benchmark for the GP. The estimator nevertheless selects its own length-scale and includes calendar month as an additional covariate, so it is not given the true generating specification. This scenario tests the fitted estimator under a data-generating process designed to be relatively well aligned with its kernel.

Nonlinear trend. A concave, decelerating trend with milder seasonality:

$$g(t) = \bar{m}(t) + 0.7 A \sin(2\pi t/12 + \phi) + r(t), \quad m(t) = 3\beta \left\{ 1 - \exp\left(-\frac{(t-2)_+}{0.22 n_{\text{pre}}}\right) \right\}.$$

Here $\bar{m}$ is the pre-treatment standardization of m . The trend approaches a plateau as its increments shrink, mimicking a bounded growth path. Unlike the first scenario, this trajectory contains nonlinear curvature that the linear kernel cannot represent on its own. The simulation examines whether the fitted GP intervals remain calibrated when extrapolation depends more heavily on recovering that curvature. The smaller seasonal amplitude keeps the main challenge in the trend rather than the seasonal component.

Saturating trend. A strong annual cycle above a leveling trend:

$$g(t) = 1.3 \beta \sin(2\pi t/12 + \phi) + 0.9 \beta \tanh(0.9 z_t) + r(t).$$

Here the annual cycle is more pronounced, while the underlying trend flattens at both ends. The scenario combines strong seasonality with a nonlinear trend, placing greater demands on both the periodic and trend components of the fitted kernel. It is intended to test whether the GP intervals remain calibrated when these features must be extrapolated together.

C.3 Design Choices

The estimator in each scenario follows the same pipeline as the application, with the length-scale selected by the variance-maximization rule, and σ^2 then estimated by marginal likelihood. No scenario gives the estimator access to the true hyperparameters, so the simulation tests the full

estimation pipeline rather than an oracle version.

The GP uses the Gaussian+Periodic+Linear kernel on the monthly index with the calendar month entered as a categorical covariate and 95% intervals. The three comparators are (i) a segmented regression carrying a linear pre-trend and an annual harmonic, (ii) **C-ARIMA**, a counterfactual ARIMA model (Menchetti et al., 2023), and (iii) **CausalImpact**, a Bayesian structural time-series model (Brodersen et al., 2015). The harmonic rather than eleven month indicators keeps the segmented fit estimable at the shortest record, and **C-ARIMA** and **CausalImpact** receive the annual period the GP is given.

D Additional Empirical Examples

Beyond the *Heller* application of the main text, we apply the method to two further settings. The All-Women Police Stations example examines a daily series with weekly seasonality and an intervention whose effects may spill over to standard police stations. The Ukraine example illustrates a more difficult setting, in which a major shock and a nonzero pre-treatment placebo flag a pre-trend.

D.1 All-Women Police Stations and Gender-based Crime

We revisit Jassal (2020)’s study of whether All-Women Police Stations (AWPS) improve women’s access to justice in India. On August 28, 2015, 20 AWPS opened simultaneously across Haryana state, a universal treatment affecting all 285 police stations. Using daily gendered crime registration rates per 100,000 population from January 2015 to August 2017 (8 months pre-treatment, 2 years post-treatment; Figure D.1), the original segmented regression analysis found a 27–49% reduction at standard stations without a system-wide increase, consistent with bureaucratic deflection as standard stations redirect women to AWPS.

We employ the Gaussian+Periodic+Linear kernel with 7-day periodicity and day-of-week covariates. Following the original study, we examine effects at both standard stations and system-wide, restricting analysis to two months post-treatment to limit the scope for concurrent events that would threaten Assumption 1. The abrupt opening date supports no anticipation for stations; Appendix D.3 takes up the residual concern of advance media coverage.

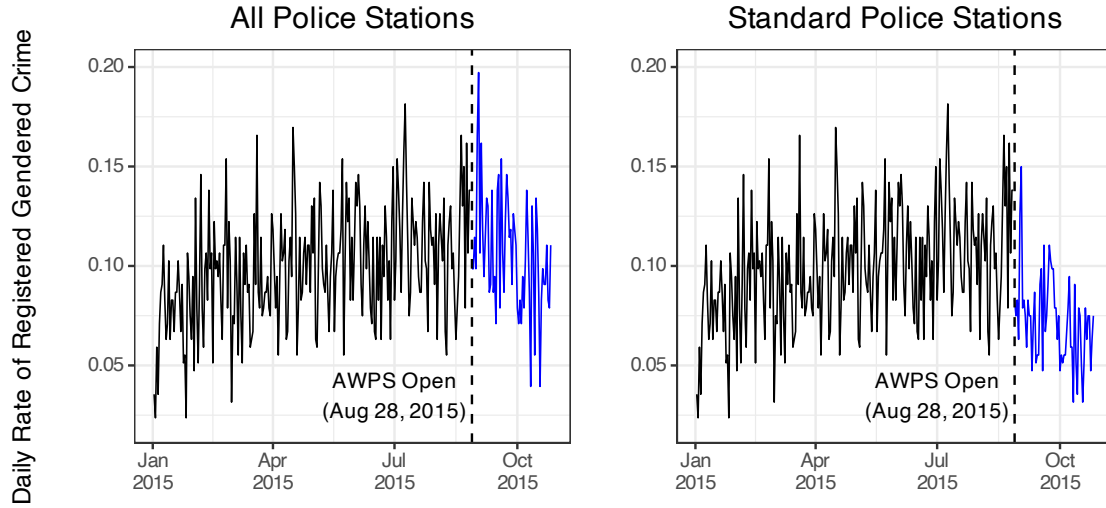


Figure D.1. Raw Outcome Data for the AWPS Application. *Note:* Daily gendered crime registration rates per 100,000 population. The vertical dashed line marks the simultaneous opening of the All-Women Police Stations (August 28, 2015).

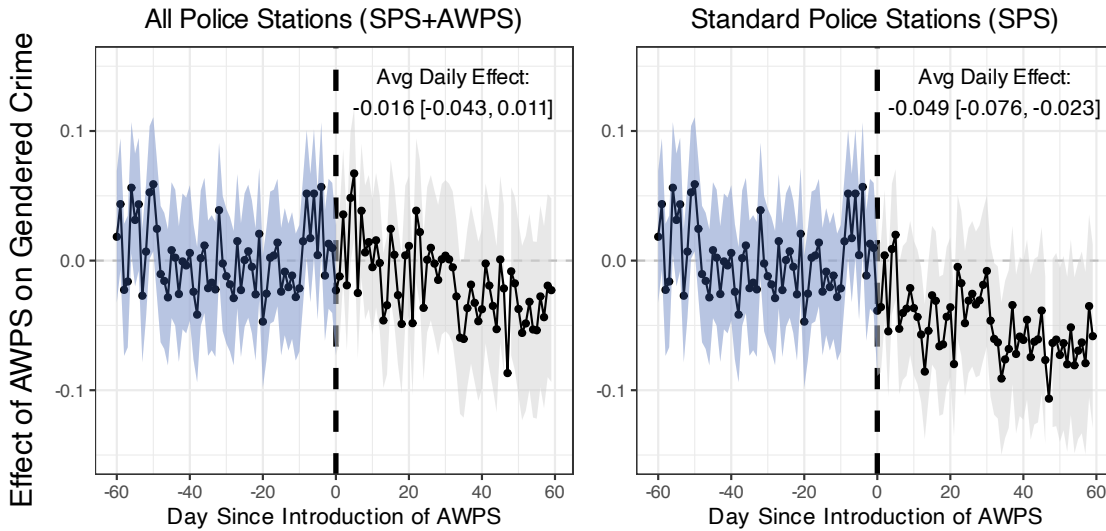


Figure D.2. Impact of All-Women Police Stations on Gendered Crime Registration per 100,000 (Jassal, 2020). *Note:* Left panel: daily treatment effects on gendered crime registrations across all police stations. Right panel: effects at Standard Police Stations (SPS) only. Pre-AWPS placebo estimates shown in blue. The introduction of AWPS did not significantly affect system-wide registration rates (left) but led to a substantial reduction at SPS (right).

Figure D.2 shows contrasting patterns. System-wide registration rates show no significant change (left panel), while standard stations exhibit substantial reductions (right panel). The average two-month effect at standard stations is -0.049 registrations per 100,000 population, with a

95% interval of $[-0.076, -0.023]$, a decrease of at least 23% from baseline. The null system-wide total is more consistent with redirection to AWPS than with net deterrence, which would lower the system-wide count. The reduction at standard stations intensifies over time, potentially reflecting increasing redirection as the new stations become established. A single level-shift estimate would obscure this gradual change.

Under the matched one-day-ahead protocol the GP covers close to the nominal rate across the 60 pre-treatment points, while the segmented alternatives fall far short (Appendix E).

D.2 The Russian Invasion of Ukraine

We reanalyze [Damann et al. \(2024\)](#), who study how the Russian invasion of Ukraine on 24 February 2022 affected gendered patterns in politicians’ public engagement. The original study fits linear ITS models with Newey-West standard errors to the Facebook posting of 469 Ukrainian politicians (81 women, 388 men) from November 2021 to June 2022 (Figure D.3). We restrict to a one-month post-treatment window to limit the scope for components of $\mathbf{U}$ taking values they never took before t_0 , and fit a separate GP per politician on pre-invasion data.

Our estimates align with the original findings: both groups increased posting after the invasion, with larger estimated effects for men (Figure D.4). The placebo check, however, reveals a slight upward pre-treatment pattern for women, suggesting that the fitted counterfactual may not fully capture their underlying trajectory. This complicates the interpretation of both the women’s estimated effect and the gender difference. If the counterfactual for women is biased upward, their effect is understated and the estimated men-minus-women difference is overstated. The placebo alone does not establish that direction of bias, so these comparisons should be interpreted cautiously.

D.3 Identifying Assumption Plausibility in the Running Examples

We assess the identifying assumptions of Section 2 in each application, focusing on potential violations that the placebo checks alone cannot resolve.

Consistency. Consistency requires a well-defined intervention and a corresponding interpretation of the potential outcomes. All three applications have a clearly dated intervention: the *Heller*

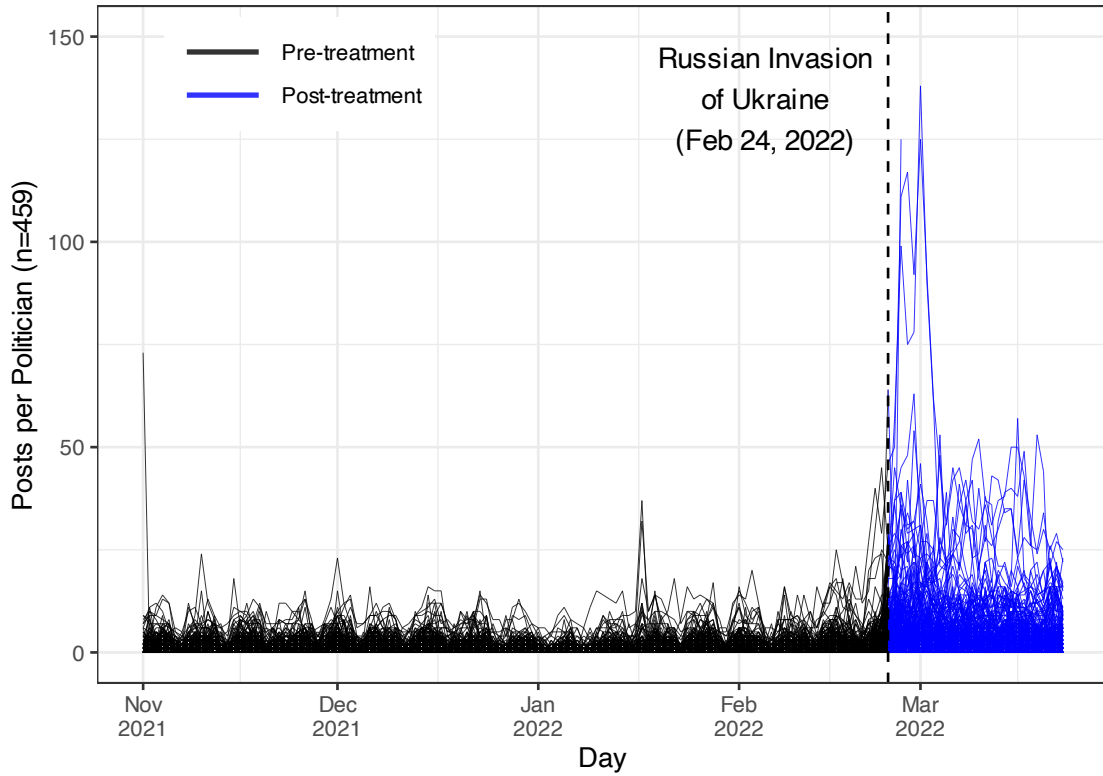


Figure D.3. Daily Facebook Posts by Ukrainian Politicians. *Note:* Daily Facebook posts of politicians from November 2021 to March 2022. Individual politician trajectories shown, pre-treatment (black) and post-treatment (blue). The vertical dashed line marks the Russian invasion on February 24, 2022. One politician is dropped for improved visualization.

ruling, the AWPS openings, and the invasion. The one subtlety is the AWPS standard stations, which are not treated directly but through the altered institutional environment around them, so their estimand is the total effect on a standard station under the new regime, inclusive of spillovers such as bureaucratic deflection. Consistency would break if “treatment” bundled several coincident changes, leaving the effect unattributable to a single intervention.

No Anticipation. The pre-treatment outcomes must be unaffected by the approaching intervention. Anticipatory behavior may begin before the formal treatment date, even when the eventual event is uncertain (Bertoli et al., 2026). Post-treatment covariates must also be unaffected by treatment. This condition holds by construction for time and calendar indicators but requires additional justification for substantive covariates.

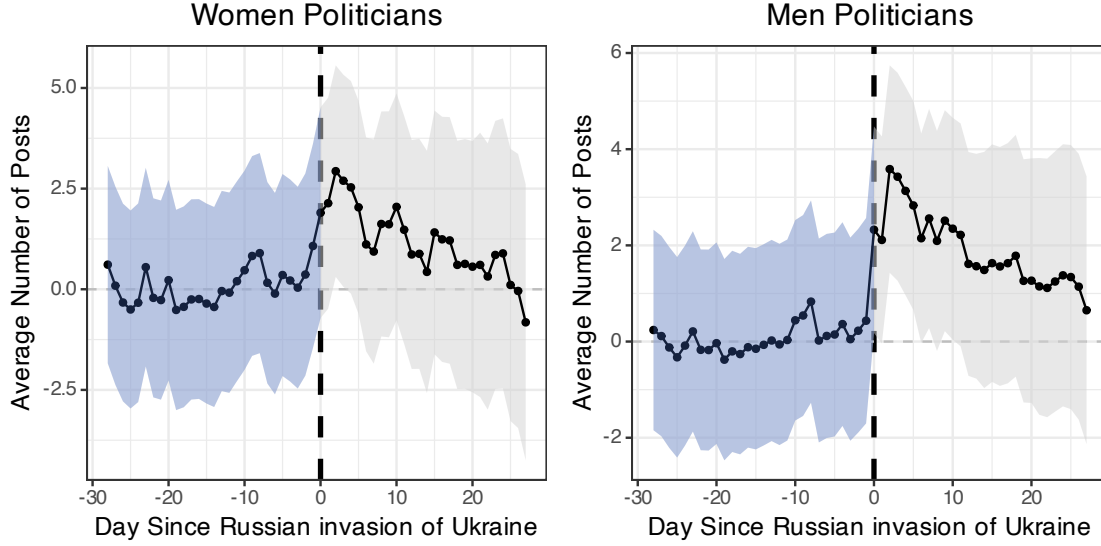


Figure D.4. Daily Average Treatment Effects by Gender. *Note:* Estimated daily treatment effects with 95% prediction intervals. Left panel shows effects for women politicians; right panel for men. A one-month pre-treatment period serves as the placebo (blue). While both groups show increased engagement, men exhibit larger effects.

For *Heller*, the March 2008 oral arguments raise a possible anticipation concern. D.C.’s strict handgun ban, however, limited the scope for preemptive legal purchases. Moving the treatment date to the oral arguments could provide an additional sensitivity check. For AWPS the abrupt opening limits the opportunity for advance redirection, although media coverage remains a concern, and the residual concern, advance media coverage, is mitigated because the eight-month pre-treatment window largely predates public awareness. For the invasion the preceding military build-up could have moved posting in anticipation, which is a potential source of pre-treatment instability that the placebo checks may detect and that we return to under Mean Sufficiency.

Mean Sufficiency (Assumption 1). The main concern is whether unobserved factors change the untreated outcome relationship after t_0 in ways that the observed covariates cannot capture (Term I, Section 4). Changes in measurement or reporting practices near the intervention may create similar problems.

For *Heller*, a major concern is the political anxiety surrounding the November 2008 presidential election, which motivates ending the analysis in October. For AWPS, concurrent changes in crime-reporting practices could affect the interpretation of the estimated effects. Unmodeled seasonal

dynamics pose a separate challenge for counterfactual prediction, which we limit by using a short two-month post-treatment window. The invasion presents the most difficult setting because its aftermath could introduce additional changes beyond the initial shock. We therefore restrict the analysis to one month and interpret the gender comparison in light of the women’s nonzero placebo estimates (Section D.2).

E Additional Results for the Empirical Applications

This section compares the GP with segmented regression using placebo checks in the *Heller* and AWPS applications. We also report cumulative national effects, residual diagnostics, and a placebo analysis of long gun purchases for *Heller*.

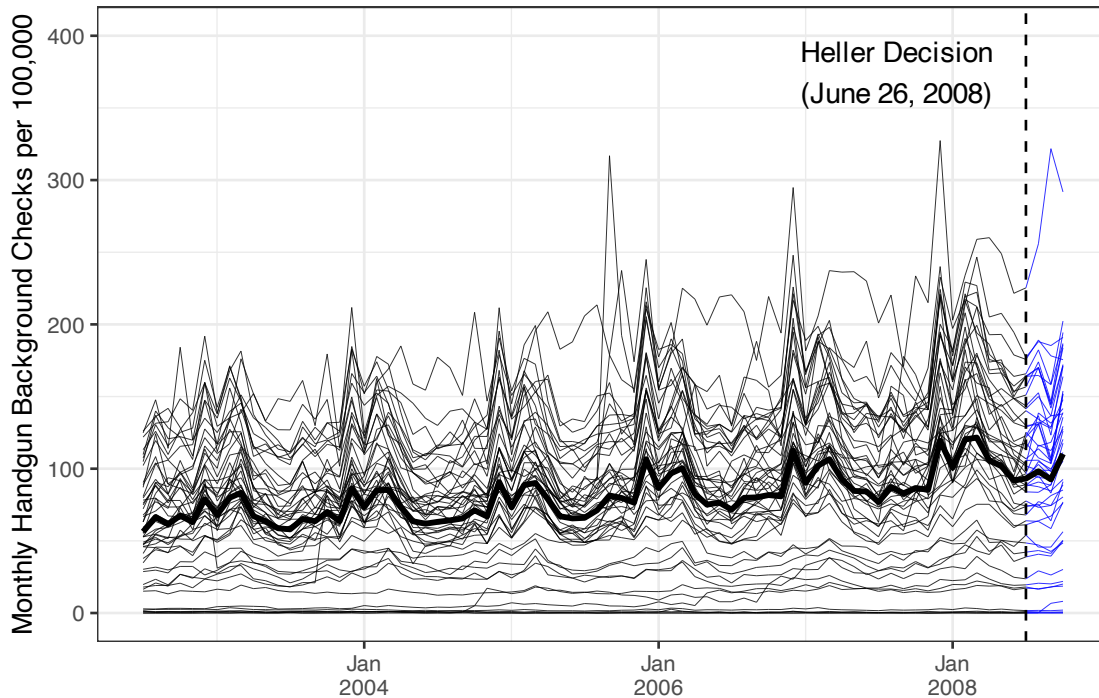


Figure E.1. Raw Outcome Data for the *Heller* Application. *Note:* Monthly handgun background checks per 100,000 population from the FBI’s National Instant Criminal Background Check System (NICS), for each state and the nationwide average. The vertical dashed line marks the *Heller* decision (June 26, 2008).

E.1 Out-of-Sample Predictive Performance with Placebo Checks: Comparison to Segmented Regression

Segmented regression is the workhorse ITS model in applied works (Bernal et al., 2017). To compare the methods under the same one-step-ahead placebo protocol of Section 3.3, we fit each model using observations through period t and predict the untreated outcome at $t + 1$. For the regression benchmark, we fit linear, quadratic, and cubic pre-treatment trends with month indicators. For each specification, the placebo effect is the difference between the observed outcome and its one-step-ahead forecast. We then record whether the corresponding 95% prediction interval contains zero.

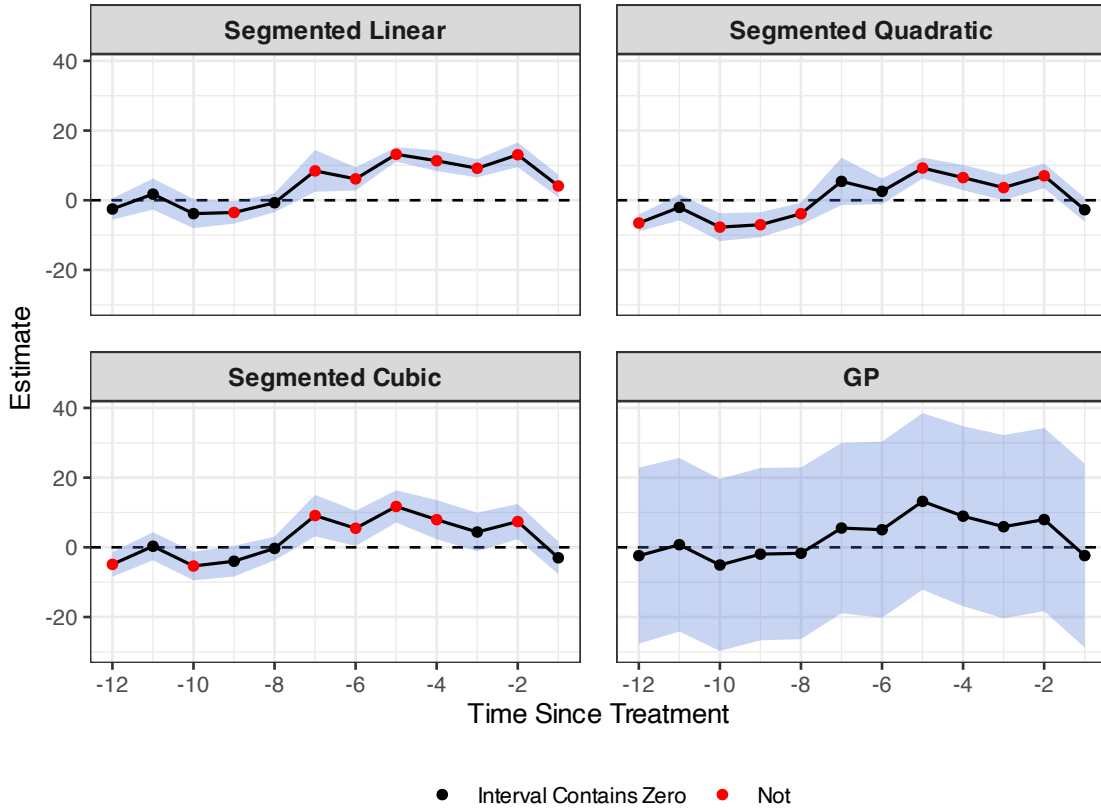


Figure E.2. Placebo Check Performance: GP and Segmented Regression for *Heller* example. *Note:* The one-month-ahead placebo check results for 1 year, comparing GP against segmented regression with linear, quadratic, and cubic time trends. Points indicate placebo treatment effects with 95% prediction intervals, colored by whether the interval contains zero (black) or not (red).

Figures E.2 and E.3 report the placebo results. For *Heller*, the GP achieves at least 95% coverage, compared with 33–42% for the segmented regressions. The latter exclude zero in 58–67% of placebo periods, as reported in Section 6. In the AWPS application, the corresponding coverage rates are 90% for the GP and 35–38% for the segmented regressions. The results are consistent

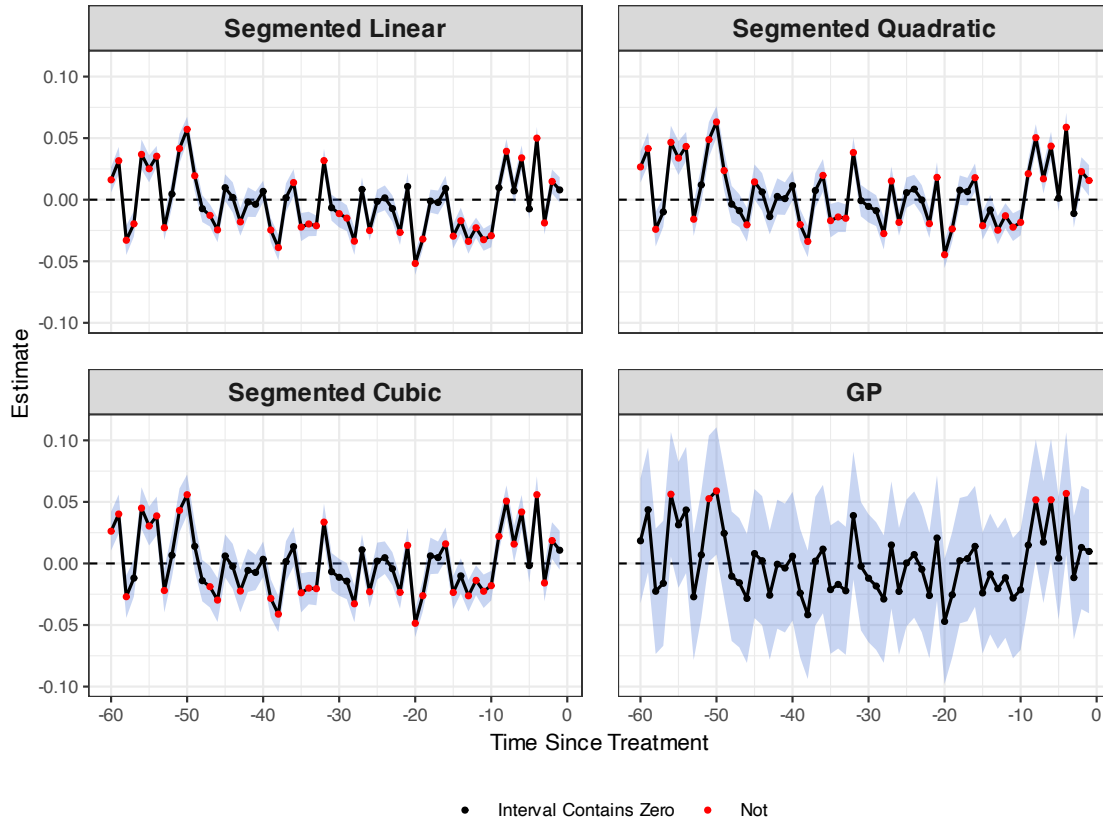


Figure E.3. Placebo Check Performance: GP and Segmented Regression for [Jassal \(2020\)](#) example. *Note:* The one-day-ahead placebo check results for 3 months, comparing GP against segmented regression with linear, quadratic, and cubic time trends. Points indicate placebo treatment effects with 95% prediction intervals, colored by whether the interval contains zero (black) or not (red).

with the importance of trend specification for extrapolation. Polynomial regressions condition their intervals on the chosen functional form, which may not capture alternative continuations of the pre-treatment series. The GP allows a more flexible set of trajectories through its combined kernel. In these placebo exercises, its intervals exclude zero substantially less often than those of the regression benchmarks.

E.2 Cumulative National Average Effects

The per-period national average effects of Section 6 can be accumulated to summarize how the response builds over the post-treatment window. Figure E.4 plots the cumulative national average effect of the *Heller* ruling on handgun background checks, together with its uncertainty interval. The band widens over the post-treatment window, reflecting growing uncertainty about the cumulative counterfactual.

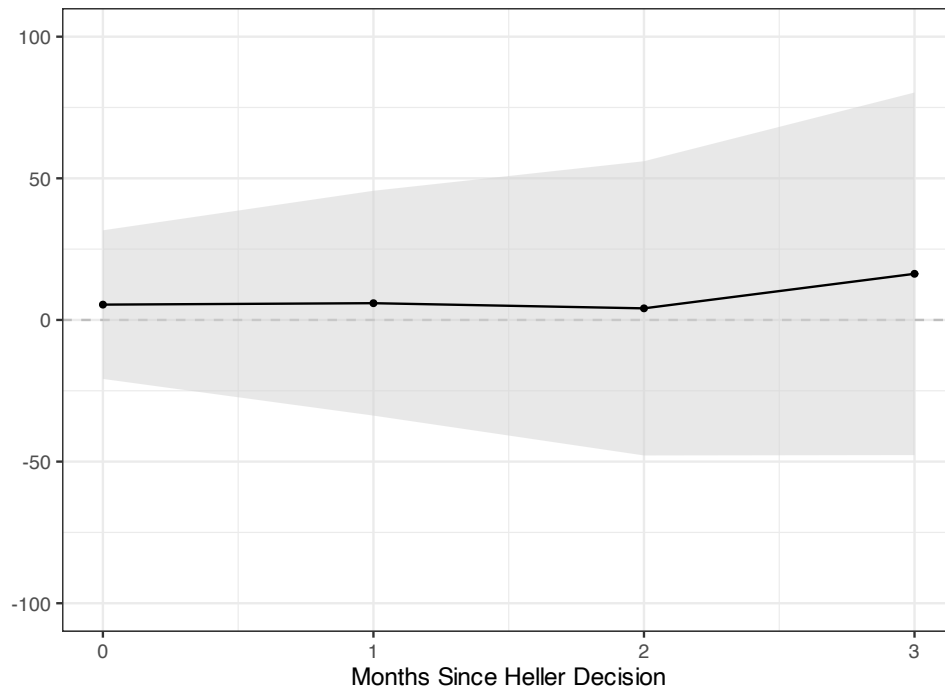


Figure E.4. Cumulative National Average Effects. *Note:* The plot shows the estimated national cumulative effects of the *Heller* ruling on handgun background checks, discussed in Section 6 of the main text.

E.3 Residual Analysis

We examine residuals from a GP fitted to the full series for each state to assess whether the model fit exhibits unusual patterns around the treatment date (Felton, 2023). Unlike the placebo checks, this is an in-sample (supplementary) diagnostic and does not evaluate out-of-sample counterfactual prediction.

Figure E.5 shows relatively stable residual variation over the pre-treatment period, without a visible increase as June 2008 approaches. A notable exception occurs in Louisiana in September 2005, shortly after Hurricane Katrina. The residual analysis provides a descriptive check on model fit, while the placebo exercises offer the more direct assessment of pre-treatment forecasting performance.

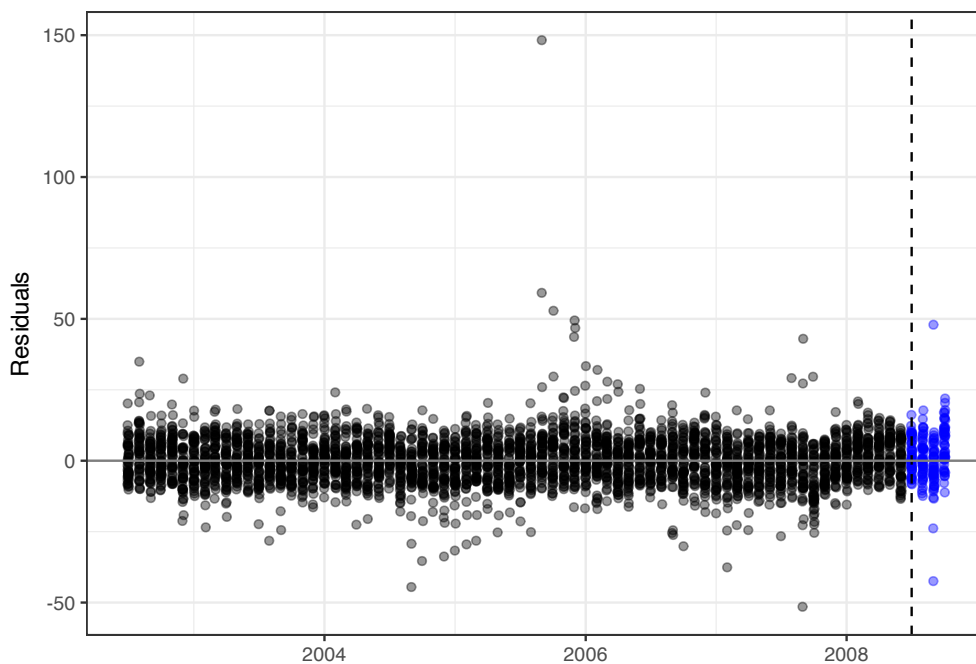


Figure E.5. Residuals from Full-Period GP Fit. *Note:* Residuals from a single GP model fitted to the entire time series for each state. Each point represents the “residuals”, i.e., difference between observed and fitted values. The vertical dashed line indicates the treatment onset.

E.4 A Pseudo-Placebo Outcome: Long Gun Purchases

Heller directly concerned handgun restrictions rather than long guns, making long gun background checks a useful placebo outcome. This makes them a placebo outcome, the cross-outcome counterpart to the temporal placebo (cf. Felton, 2023). Figure E.6 runs the same analysis for long gun checks that Figure 4 runs for handguns. The state-specific effects cluster around zero with no clear geographic or political pattern, and the national effects fluctuate around zero without trend. The absence of a comparable change in long gun checks provides a complementary placebo result for the handgun analysis.

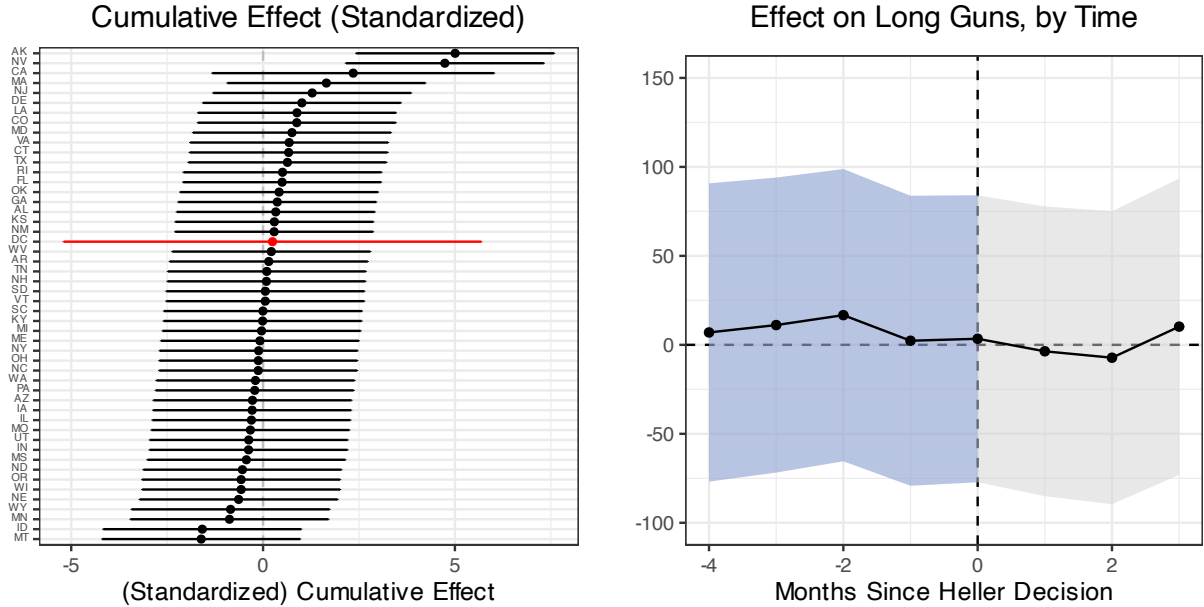


Figure E.6. Effects of Heller on Long Gun Background Checks. *Note:* State-specific (standardized) cumulative treatment effects (left panel) for the four-month period and national average treatment effects over time ($\hat{\tau}_{t*}$, right panel) for long gun background checks following *Heller*. In contrast to handgun purchases, long gun checks show no systematic response to the decision, with effects centered near zero across both states and time periods.

F Details on Covariate Processing and Kernel

Covariate scaling. We center the continuous covariates and the outcome and scale each to unit variance within every unit, so the kernel reads all continuous coordinates on a common scale and its behavior does not depend on the units in which variables were recorded (Doudchenko and

Imbens, 2016; Hainmueller and Hazlett, 2014). Posterior means and intervals are then mapped back to the original outcome scale, so estimated effects and their uncertainty are reported in the natural units of the outcome. Categorical coordinates enter the same distance through a separate convention.

Categorical covariates. Categorical covariates, such as the month and day-of-week indicators used in our applications, need different treatment from the continuous case (Johannemann et al., 2019). Ordinary regression drops one level to avoid collinearity, but a kernel reads the covariates through distances, and dropping a level would make those distances asymmetric across categories, since the omitted level would sit at the origin and appear closer to every other level than the remaining levels are to one another. We instead one-hot encode without a reference level, writing $\mathbf{e}(\ell)$ for the indicator vector of level ℓ ; any two distinct levels are then equidistant, $\|\mathbf{e}(a) - \mathbf{e}(b)\|^2 = 2$, the symmetry appropriate for unordered categories with no natural baseline. We then rescale each indicator by $\sqrt{1/2}$, so that a mismatch on a categorical covariate contributes

$$\left\| \sqrt{\frac{1}{2}} \mathbf{e}(a) - \sqrt{\frac{1}{2}} \mathbf{e}(b) \right\|^2 = \frac{1}{2} (1 + 1) = 1, \quad a \neq b. \quad (\text{F.1})$$

Thus, a categorical mismatch contributes one unit of squared distance, the same as a one-standard-deviation difference in a standardized continuous covariate (Hartman et al., 2025). This scaling keeps the relative contribution of categorical and continuous covariates explicit.

Length-scale hyperparameter b . We select b using the variance-maximization rule of Hartman et al. (2025), which maximizes the variance of the off-diagonal entries of the kernel matrix. To assess sensitivity to this choice, we re-estimate the treatment effects over a broad range of b values. Figures F.1 and F.2 show that the large positive D.C. estimate, the near-zero national average, and the overall distribution of state-specific point estimates remain stable across the values examined.

Period hyperparameter p . The period p determines how often a pattern repeats. The periodic kernel assigns maximum covariance to time points separated by integer multiples of p . We set p according to the natural cycle of the data, such as $p = 12$ for monthly data with annual seasonality, $p = 52$ for weekly data with annual seasonality, or $p = 7$ for daily data with weekly cycles.

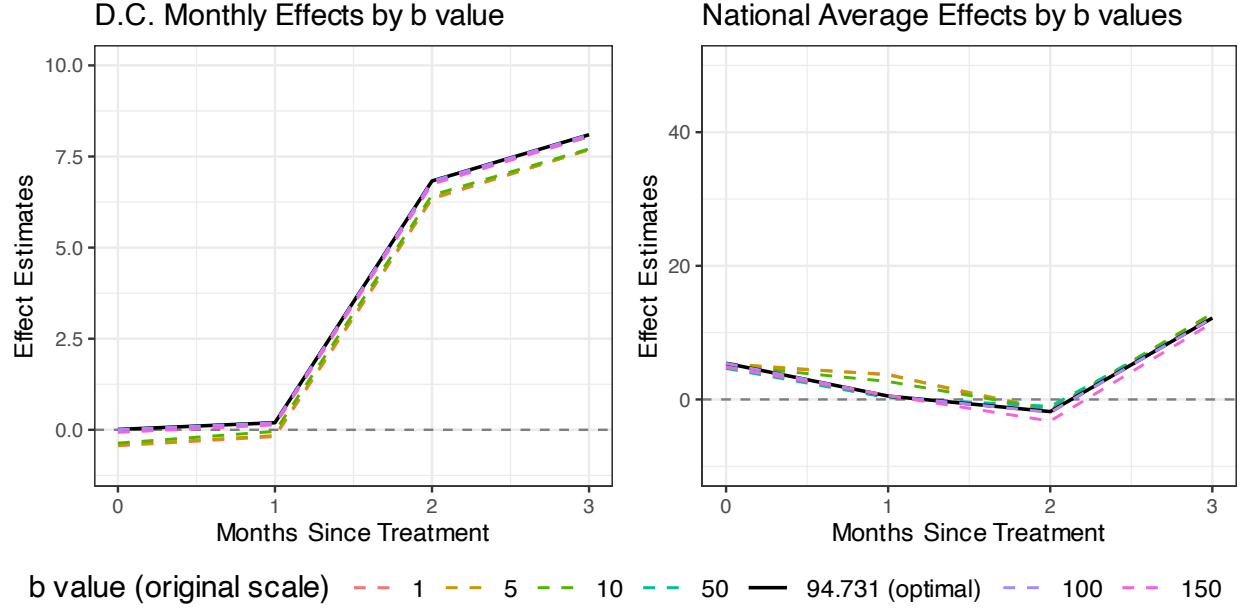


Figure F.1. Sensitivity of treatment-effect estimates to the length-scale b . *Note:* The left panel shows monthly effects for D.C., and the right panel shows national average effects. The solid black line marks $\hat{b}$ selected by the variance-maximization rule. Prediction intervals are omitted for visual clarity.

When the period is unknown, it can be selected by detrending the pre-treatment series, computing its periodogram, and converting the frequency of maximum power to a period ([Bloomfield, 2004](#)).

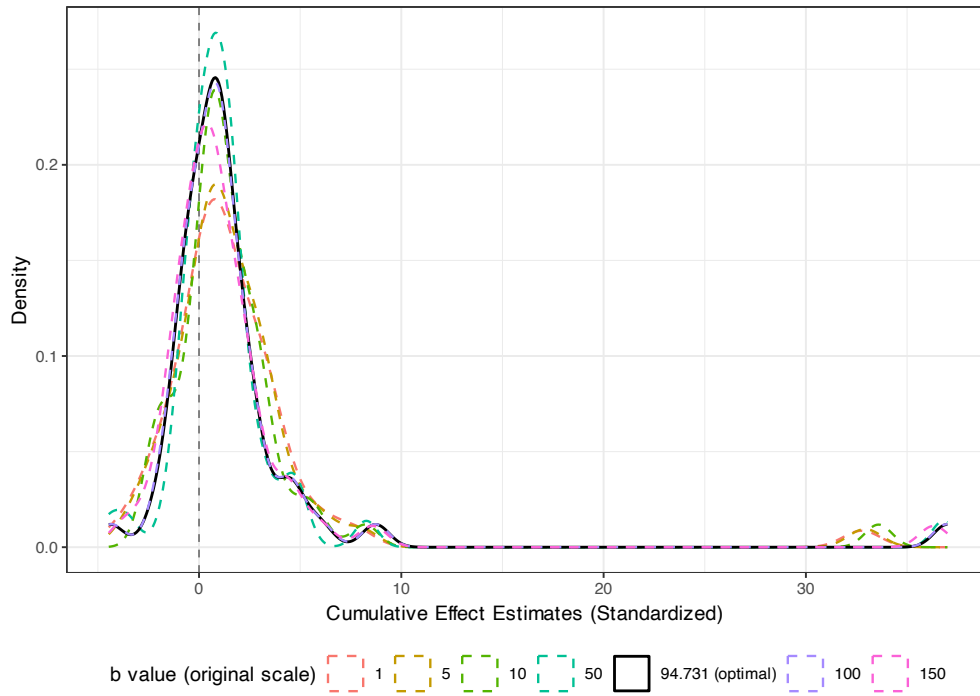


Figure F.2. State-specific cumulative treatment-effect estimates across length-scale values. *Note:* The distribution obtained with the selected length-scale ($\hat{b} = 94.73$, solid black line) is compared with those obtained using alternative values (dashed colored lines). Prediction intervals are omitted for visual clarity.

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